

# XR Performance

## Expanding Experimental Media Through Spatial Computing and Real-Time Interaction

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### Abstract

At LSU, a constellation of courses has been developed around a new XR Studio, which features a large-scale LED wall as the centerpiece for curricula focused on hybrid media in performance and filmmaking. One such course, XR Performance, is a cross-disciplinary offering at the intersection of extended reality (XR), experimental media, and performance studies. Designed for students from a wide range of academic backgrounds, the course explores emerging forms of storytelling, spatial computing, and real-time engine interaction, including hands-on experience with motion capture. In addition to XR Performance, courses like XR Horror Production and Intro to Virtual Production focus on virtual production workflows for filmmaking, immersive sound design, and traditional skills such as prop making. A newer offering, Digital Twinning, emphasizes 3D scanning and the pairing of real-world subjects, objects, and robotics with virtual counterparts, using the LED wall as a platform for spatial interaction. A novel aspect of this curriculum is the use of consistent XR tools and software—such as Unreal Engine and motion capture systems—across all courses, despite their varied topics. Together, these offerings challenge the boundaries between digital and physical space while enabling students to critically examine how XR reshapes storytelling, audience engagement, and interdisciplinary collaboration.

### CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; **Virtual reality**; **Interaction design**; • **Applied computing** → **Media arts**; **Performing arts**; • **Computing methodologies** → **Real-time simulation**; **Motion capture**; • **Social and professional topics** → **Computing education**.

### Keywords

XR Performance, Immersive Storytelling, Virtual Production, Mixed Reality, Digital Twinning, Interactive Media, Motion Capture

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## 1 Introduction

As the boundaries between physical and digital spaces continue to blur, extended reality (XR) technologies have emerged as powerful tools for experiential learning and creative exploration. *XR Performance*, an interdisciplinary course at Louisiana State University, positions XR not just as a medium for visual immersion, but as a framework for live performance, genre experimentation, and embodied interaction. Students from diverse academic backgrounds use real-time game engines, motion capture, and LED walls to create responsive works that blend narrative, technology, and audience engagement.

This paper presents a pedagogical model grounded in spatial computing, embodied interaction, and collaborative project-based learning. The framework emphasizes hands-on experimentation, shared technical workflows, and interdisciplinary collaboration to help students explore XR as both a technical skillset and a creative medium. This paper presents case studies from three courses—*XR Performance*, *XR Horror Production*, and *Digital Twinning*—to demonstrate how this shared foundation can support varied educational goals, from immersive theater to genre filmmaking and real-time simulation.

## 2 XR Horror Production & Virtual Production

*XR Horror Production* is a course centered on an exploration of virtual production through the lens of horror. This course merges emerging technology with the corn syrupy tradition of the horror genre, emphasizing the intersection of technical skills and creative

storytelling. In the first half of the semester, students gain experience in key roles within the film production pipeline through structured projects, learning the fundamentals of XR technologies like LED walls and Unreal Engine. In the second half of the course, students focus on collaborative filmmaking in a large-scale production that reflects horror's capacity to address cultural anxieties. The course structure reinforces project-based learning and cross-functional collaboration, with students rotating through roles in a real-time production pipeline while working within a shared technical framework.

### 3 XR Performance

Performance and live events such as concerts, opening ceremonies, theatrical productions, and other spectacles are using light, sound, and emerging display technologies to extend reality in rapidly evolving ways. This course explores the intersection of electronic music, visualization, performer, and audience. The class focused on creating graphics, lighting, and visual illusions that respond and interact in real time with music, sound, and performance on a large LED wall in LSU's new XR Studio environment. Experimental and traditional tools for content creation were included in the production pipeline, such as Unreal Engine, Notch, and TouchDesigner. Other strategies for interaction and response—such as motion capture, sensors, and computer vision—were also introduced. The course was project-based and entailed a series of lessons that paved the way for original XR performances that combined physical stagecraft with real-time electronic media. Students explored XR as a performative and spatial medium, engaging deeply with embodied interaction and collaborative creation.

### 4 Digital Twinning

*Intro to Digital Twinning* explores the creation of real-time virtual counterparts to physical objects, spaces, and people using tools like Unreal Engine and NVIDIA Omniverse. Students build interactive environments connected to devices such as robotic arms or 3D-scanned representations of themselves, interacting with them through immersive interfaces like augmented and virtual reality. These XR-based workflows allow students to engage with complex systems spatially rather than abstractly, providing more intuitive understanding. The course also introduces hybrid pipelines that integrate 3D scanning, sensor data, and AI-driven tools such as neural radiance fields (NeRFs), offering students both technical fluency and a conceptual foundation for working with responsive, real-time environments. The shared toolset also prepares students to carry their skills into other contexts, from horror filmmaking to immersive performance.

### 5 Conclusion

The LSU XR Studio has proven to be a transformative space for interdisciplinary education, bridging technology, performance, and experimental media in ways that redefine creative expression. Through courses such as *XR Performance*, *XR Horror Production*, and *Digital Twinning*, students gain hands-on experience with real-time rendering, virtual production, and motion capture while exploring the broader implications of extended reality on storytelling and audience engagement. These courses are pedagogically intertwined,

allowing students to transfer skills across contexts—for example, applying 3D scanning and spatial programming techniques from *Digital Twinning* to a horror short in *XR Horror Production*, or incorporating scanned assets into a live *XR Performance* on the LED wall. By integrating emerging tools like Unreal Engine, NVIDIA Omniverse, and motion-tracking systems into a shared curriculum, the XR Studio fosters an environment where students from diverse disciplines can collaborate and explore the intersections of digital and physical media. Collectively, these courses demonstrate a pedagogical framework that combines spatial and embodied learning, shared technical platforms, and interdisciplinary collaboration. By sharing our project-based approach and examples of student work, we aim to contribute to the broader conversation around how XR can support interdisciplinary learning, creative practice, and technical skill development in higher education.